

LITTLE FLATLAND ISLAND, ON, Canada

WMO: 714680

Lat: 49.690N

Lon: 88.306W

Elev: 856

StdP: 14.25

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.7	-16.3	-26.0	1.3	-20.6	-21.3	1.8	-16.2	32.2	9.9	29.5	7.4	8.2	290	0.710

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.5	76.9	67.4	74.3	66.0	71.7	64.4	69.1	74.8	67.3	72.4	65.5	70.2	6.6	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.0	102.5	72.3	65.2	96.1	70.4	63.6	90.8	68.6	33.7	74.8	32.3	72.7	30.9	70.1	75.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min		Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
30.1	26.8	23.7	DB	-25.8	83.7	6.4	3.1	-30.5	85.9	-34.2	87.7	-37.8	89.4	-42.5	91.6	
			WB	-26.0	72.1	6.0	2.3	-30.3	73.8	-33.9	75.1	-37.2	76.4	-41.6	78.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	37.0	5.9	6.8	18.7	31.3	43.3	54.8	63.7	64.0	56.1	43.6	31.6	22.0	
	DBStd	21.67	13.33	12.81	12.35	8.61	6.57	6.31	5.24	5.00	6.59	7.08	8.65	9.26	
	HDD50	6023	1367	1210	971	560	225	25	0	0	21	224	553	867	
	HDD65	10328	1832	1630	1436	1010	672	309	88	78	273	665	1003	1332	
	CDD50	1277	0	0	0	0	19	170	425	435	203	25	0	0	
	CDD65	104	0	0	0	0	0	4	47	48	5	0	0	0	
	CDH74	258	0	0	0	0	1	14	132	107	4	0	0	0	
	CDH80	18	0	0	0	0	0	1	11	6	0	0	0	0	
Wind	WSAvg	10.4	10.8	9.7	9.2	8.9	8.3	7.6	8.4	9.6	11.0	13.2	13.7	13.9	
Precipitation	PrecAvg	34.6	1.9	1.3	1.7	2.6	3.5	3.9	3.8	3.5	4.1	3.6	2.7	2.1	
	PrecMax	41.3	4.4	3.3	3.3	7.7	8.1	8.0	6.7	5.4	8.5	6.6	5.2	3.7	
	PrecMin	25.4	0.4	0.4	0.4	0.7	1.9	1.8	2.2	1.7	1.7	1.3	1.0	1.0	
	PrecStd	4.0	0.7	0.7	0.6	1.3	1.2	1.4	1.1	1.0	1.5	1.3	1.0	0.7	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.1	37.7	45.8	54.3	67.9	75.8	81.3	80.6	73.7	62.9	52.3	41.6	
		MCWB	33.5	35.1	41.3	44.1	54.7	62.9	69.8	69.3	64.8	56.2	47.7	39.0	
	2%	DB	32.9	34.6	40.5	48.0	61.6	71.4	77.4	77.0	70.5	59.1	48.6	38.6	
		MCWB	32.2	33.3	36.6	40.6	50.9	61.9	67.7	68.1	63.7	54.4	45.8	36.8	
	5%	DB	30.5	31.9	37.6	44.4	56.7	68.2	74.7	74.3	67.5	56.2	45.8	36.5	
		MCWB	29.7	30.8	35.1	38.5	49.0	60.2	66.6	66.4	62.1	52.6	43.2	34.9	
	10%	DB	26.2	27.2	35.1	41.2	52.8	64.6	72.0	71.6	64.8	53.6	43.3	34.4	
		MCWB	25.4	26.2	33.6	36.9	46.9	58.3	64.9	64.9	60.8	50.4	41.1	33.2	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.0	35.7	42.7	45.0	57.1	65.7	71.9	71.9	66.9	57.9	48.8	39.4
			MCDB	34.6	36.9	44.3	51.7	63.6	72.2	78.1	77.7	71.1	61.6	50.9	40.9
2%		WB	32.3	33.4	37.2	41.3	52.8	63.0	69.3	69.3	64.6	55.3	45.9	37.2	
		MCDB	32.9	34.6	39.8	46.5	58.5	69.3	75.5	74.7	68.3	58.0	48.1	38.6	
5%		WB	29.6	30.8	35.2	39.0	49.8	61.0	67.4	67.7	63.1	53.1	43.6	35.2	
		MCDB	30.5	31.9	37.0	43.6	55.2	66.5	72.9	72.6	66.5	55.8	45.5	36.1	
10%		WB	25.4	26.2	33.7	37.1	47.2	58.8	65.6	66.0	61.3	50.6	41.2	33.3	
		MCDB	26.1	27.1	35.2	40.7	51.9	63.5	70.6	70.5	64.4	53.2	43.3	34.3	
Mean Daily Temperature Range	5% DB	MDBR	13.8	17.6	17.5	14.9	14.9	14.8	13.9	12.5	11.1	8.2	7.8	8.8	
		MCDBR	12.9	15.4	16.0	18.0	22.1	19.8	17.5	16.0	14.1	12.0	9.4	8.6	
		MCWBR	12.5	14.1	12.4	11.4	13.4	10.9	9.5	8.9	8.4	8.4	7.5	8.0	
	5% WB	MCDBR	13.1	15.2	14.0	16.5	19.9	17.3	16.4	14.9	12.7	11.3	9.3	8.6	
		MCWBR	12.8	14.1	11.1	10.9	13.2	10.6	9.7	9.0	8.1	8.3	7.8	8.2	
Clear-Sky Solar Irradiance	taub		0.246	0.266	0.299	0.343	0.353	0.370	0.390	0.367	0.350	0.308	0.283	0.259	
	taud		2.307	2.266	2.305	2.372	2.429	2.415	2.388	2.448	2.492	2.575	2.429	2.279	
	Ebn at Noon		260	281	289	287	287	282	274	276	270	266	244	233	
	Edh at Noon		21	29	33	35	35	36	36	32	28	22	19	19	
All-Sky Solar Radiation	RadAvg		392	768	1186	1529	1661	1785	1778	1506	1072	623	318	265	
	RadStd		49	65	103	143	93	135	124	124	86	93	36	38	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page